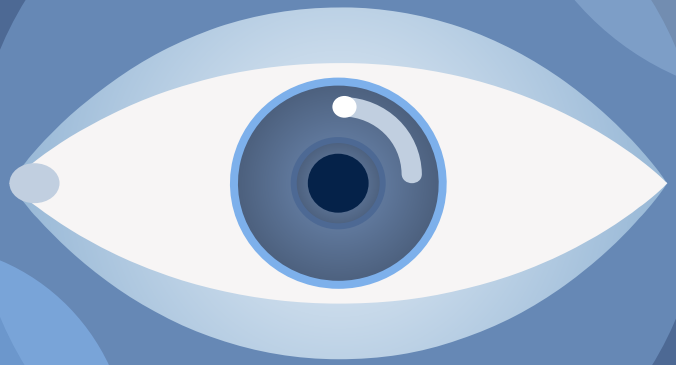
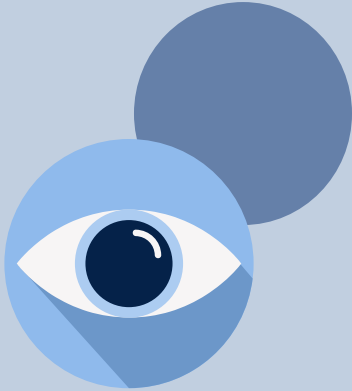
Glaucoma

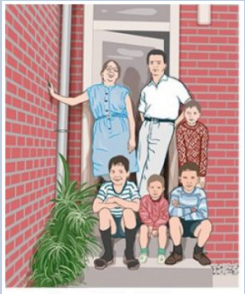
Presenting:
Yuval, Shahar and Tair



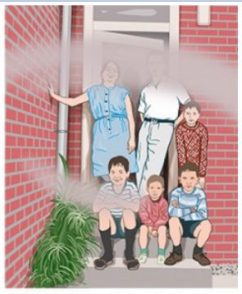
What is Glaucoma?



- An optic degenerative disease that is expressed in damage to the peripheral nervous system and causes loss of retinal ganglion cells.
- Glaucoma is not easily identifiable and can lead to irreversible vision loss.
- A leading cause of irreversible blindness and a decrease in quality of life.
- Glaucoma is often caused by increased intraocular pressure



Normal field of vision



Field of vision with early glaucoma

Field of vision with advanced glaucoma

Field of vision with severe glaucoma

By 2040

110 million

Two Types of Glaucoma

Primary glaucoma

01 Two types:
Open-angle or closed angle

02 No cause or reason that can be identifiable

Secondary glaucoma

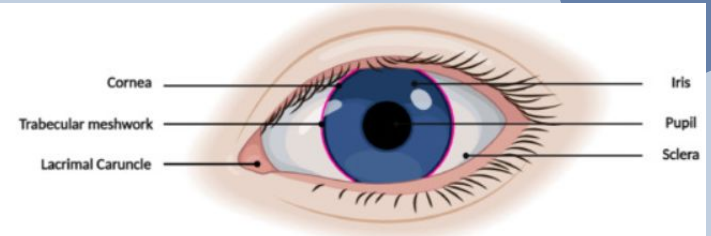
01 Two types:
Open-angle or closed angle

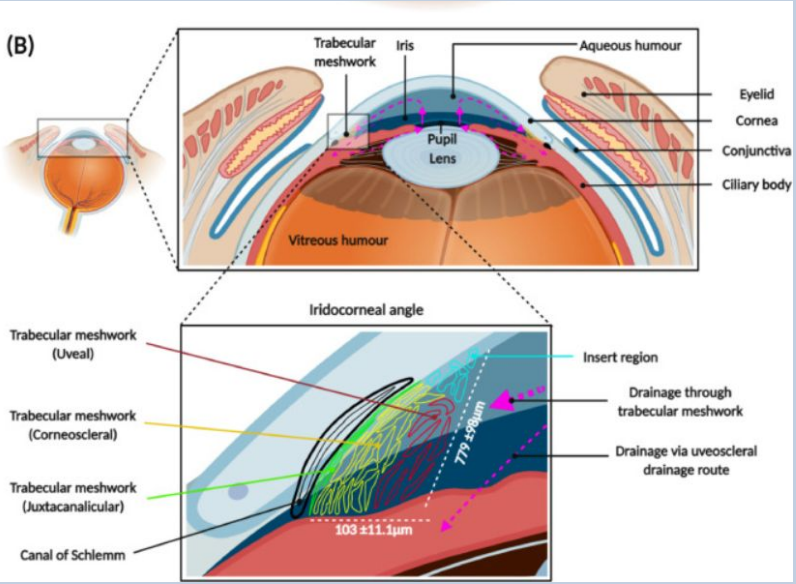
02 Caused by Intraocular pressure that increased, which causes damage to the optic nerve

The incidence of blindness is higher in closed-angle glaucoma although closed-angle glaucoma is less common than open-angle glaucoma

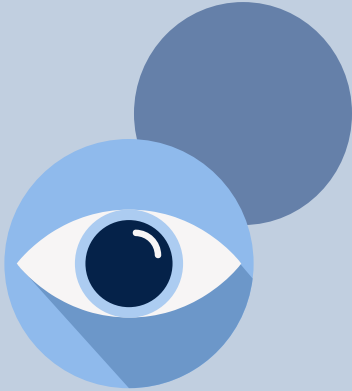
The structure of the eye

- A **ganglion cell** - a type of nerve cell found in the retina that gets visual and passes it on to the brain through the optic nerve.
- **Astrocytes** are the main cell type in the optic nerve head and regulate the expression of different cell surface molecules
- The **Trabecular Network** (TM) is a spongy tissue composed of a group of tiny channels through which most of the fluid in the eye drains. It has the ability to contract and can control eye pressure





Astrocytes



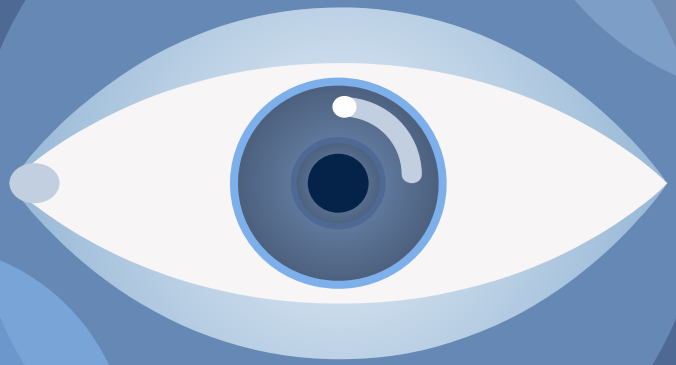
- The astrocytes regulate the expression of different cell surface molecules that play important roles in cell-cell recognition and cell adherence to the sheets.
- In glaucoma, astrocytes play a central role in causing the reshaping of the extracellular matrix of the optic nerve head, growth factor synthesis, and mediation with other cells.
- Astrocytes may respond to changes in intraocular pressure in glaucoma and thus underlie the degeneration of ganglion cells in the retina.

The impact of the disease

01 The Eye Pressure

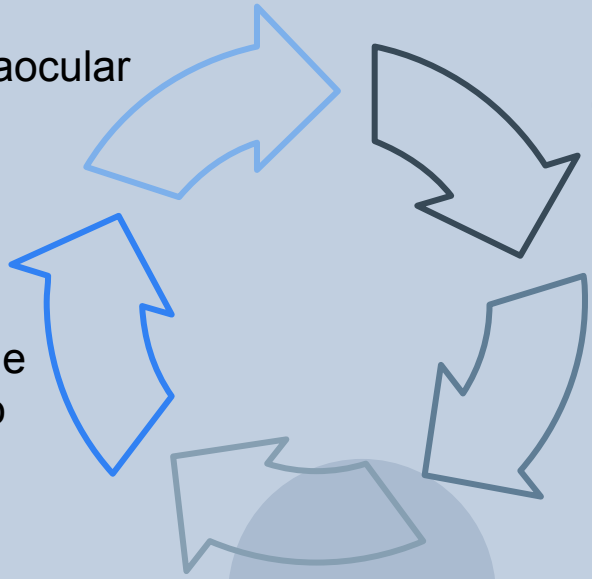
02 Impact on the two-way signaling

03 Impact on the integrators



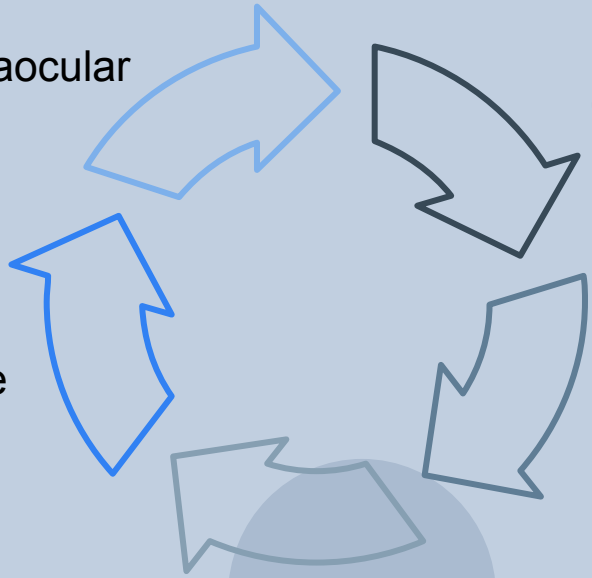
The Eye Pressure

First Impact



The Eye Pressure

Second Impact



Abnormal Intraocular pressure

Levels of extracellular calcium ions increase

An increase in TGF-beta cytokine levels

Increasing production and subsidence of extracellular matrix proteins in TM cells

A block the excretion of the eye fluids

Impact on the two-way signaling

The changes most often associated with glaucoma include:

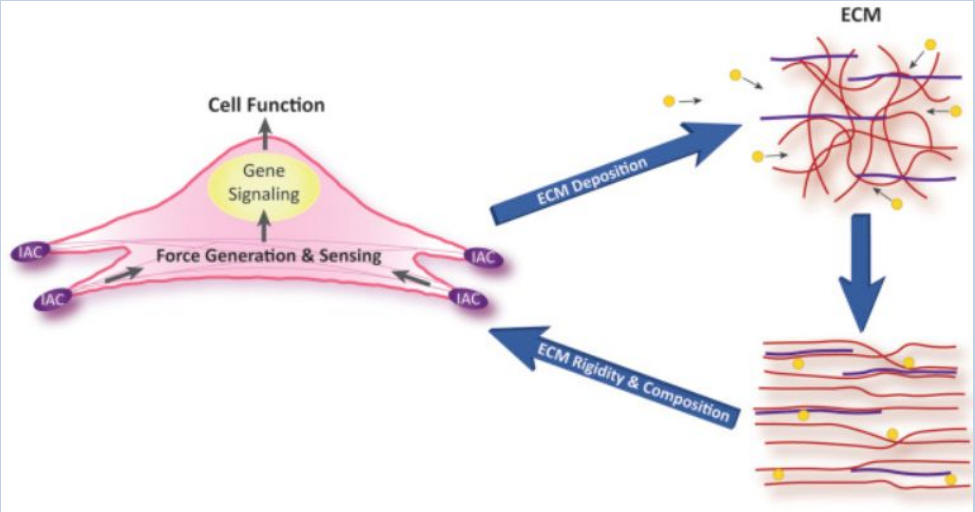
01 Changes in contraction properties of the actomyosin of the Trabecular Network (TM)

02 Changes of the extracellular matrix (ECM) by the amount and type of proteins that are produced.

All these changes are interrelated and result from a two-way signal between the ECM and cells

Impact on the two-way signaling

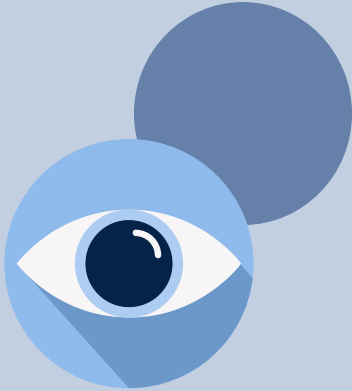
Changes in the biological properties of the cell



Changes in the extracellular matrix (ECM)

Changes in the signals captured by the Integrin (IAC)

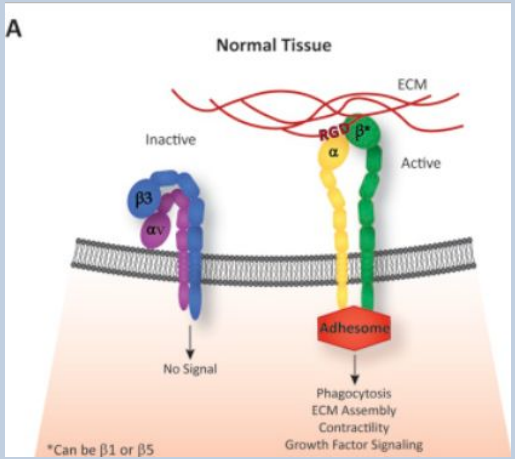
Impact on the Integrins



- The most involved receptors in two-way communication are the Integrins.
- The major ligands for integrins are ECM proteins.
- The integrins play an important role in the signal transfer between the ECM and the cells

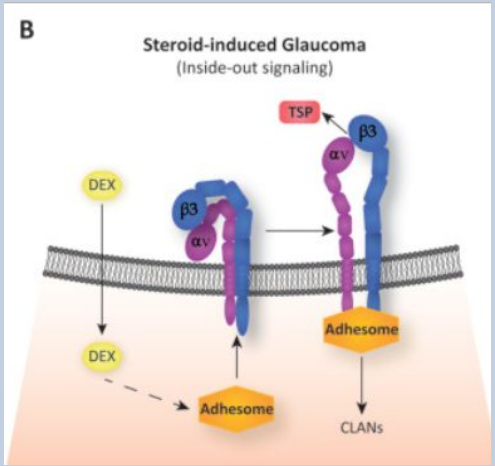
Impact on the Integrins

Normal Tissue



- Integrins in an inactive-bent structure and active state.
- In the active state, its structure enables interactions.
- In this configuration, biological processes can be regulated

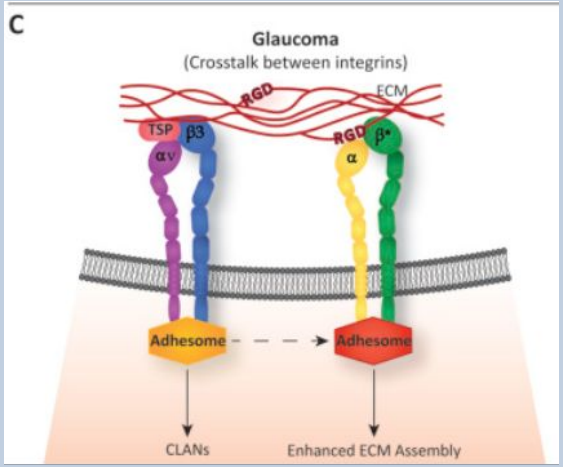
Steroid induced Glaucoma



- Inactive Integrins in TM can be activated by DEX.
- The change in configuration allows it to bind to a ligand such as TSP.
- inside-out signaling causes the formation of the CLAN and therefore may be involved in glaucoma caused by steroids.

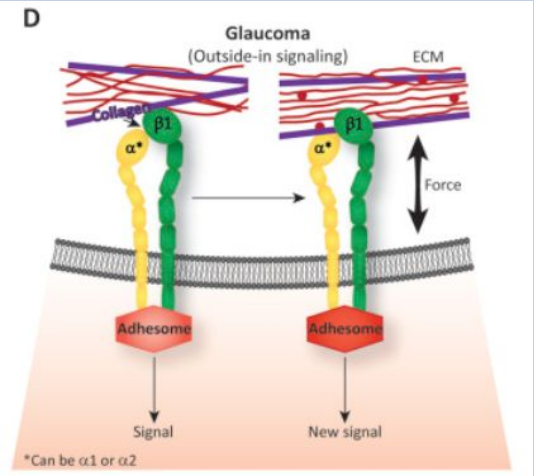
Impact on the Integrins

Crosstalk between integrins



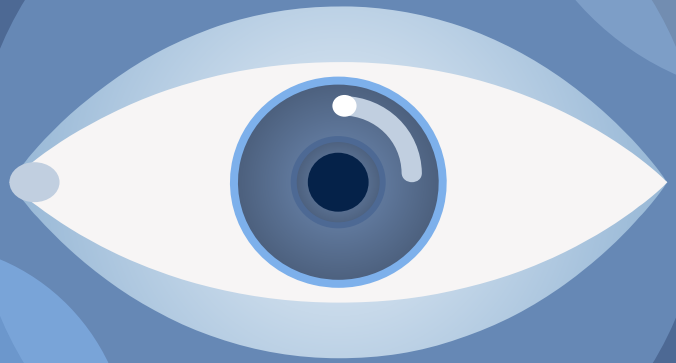
- Cross-linking between active Integrins and other Integrins can affect other signaling events in TM cells and trigger glaucoma.
- The cross-linking can cause reduced phagocytosis and the formation of a CLAN and increase in the amount of complex fibronectin.

Glaucoma



- Synthesized proteins in ECM can bind to other integrins and initiate the formation of new signaling events.
- Presence of Integrins on the cell surface of TM cells indicates participation in glaucoma pathophysiology

Treatment



Treatments

Medicines

Reduces intraocular pressure using various medications

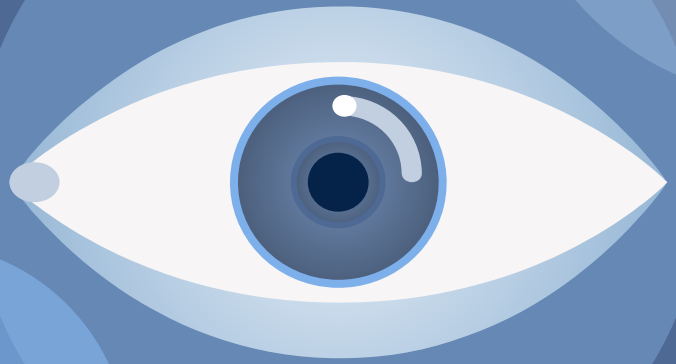
Laser treatment

Reduces intraocular pressure by causing biological changes to the Trabecular network

Surgical

During this process, a small portion of the trabecular network or adjacent cornea tissue removed

Teleglaucoma

Allows identification of glaucoma and those at risk of developing the disease by transmitting high-resolution stereoscopic fundus images electronically

Thank you for listening